CN Practical 10: Implement Classful Addressing in a Given Network

Aim: To identify the range of IP addresses in different classes and justify the selection of suitable IP address classes for a given network.

Theory:

IPv4 addresses are 32-bit logical addresses traditionally divided into five classes: A, B, C, D and E. This method is called classful addressing because the network and host portions are determined by the address class. Classes A, B and C were used for unicast addressing, Class D was assigned to multicast, and Class E was reserved for experimental purposes.

Class A was designed for large networks, Class B for medium-sized networks and Class C for smaller networks. Each class has a default subnet mask. Although classful addressing is important for understanding IP addressing, modern networks primarily use Classless Inter-Domain Routing (CIDR), which allows flexible prefix lengths.

IP address classes:

Class	First-octet range	Default mask	Purpose
A	1–126	255.0.0.0	Large networks
B	128–191	255.255.0.0	Medium networks
C	192–223	255.255.255.0	Small networks
D	224–239	Not applicable	Multicast
E	240–255	Not applicable	Reserved/experimental

Note: 127.x.x.x is reserved for loopback, and some address ranges have special purposes.

Procedure:

1. Open Cisco Packet Tracer or refer to an IPv4 address table.
2. Select sample IP addresses from Classes A, B, C, D and E.
3. Examine the first octet of each address.
4. Identify the class using the first-octet range.
5. Write the default subnet mask for Classes A, B and C.
6. Identify the uses of Class D and Class E addresses.
7. Consider a network's size and determine which traditional class would be suitable.
8. Record the address ranges, subnet masks and applications in a table.

Observation:

The IP addresses were classified according to their first octet. Classes A, B and C have different default network portions and were traditionally associated with different network sizes. Class D is used for multicast, while Class E is reserved for experimental purposes.

Conclusion:

The different classes of IPv4 addresses and their ranges were studied. Their default subnet masks and traditional applications were identified. The practical demonstrated the fundamentals of classful addressing and the selection of IP addresses for networks.